MAMIDI HEALTH INC.

Whole Genome Sequencing - Research Variant Interpretation Report

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Proband Identifier:	**WGS_EX2312012**	**Sample Display Name:**	WGS_EX2312012_HGWCNDSX7
Sample Type:	Constitutional / Germline DNA	**Sequencing Assay:**	Whole Genome Sequencing (WGS)
Reference Assembly:	GRCh38 / hg38	**Analysis Protocol:**	Rare Disease Singleton Workflow

2. CLINICAL INDICATION & PHENOTYPIC PROFILE

The proband is an infant presenting with an early soft tissue malignancy in the setting of severe congenital growth restriction, prematurity, and neonatal renal calcification. Documented clinical features mapped to Human Phenotype Ontology (HPO) terms comprise:

HPO Term	HPO ID	Clinical Presentation and Notes
Rhabdomyosarcoma	HP:0002859	Primary oncological event: Malignant soft tissue tumour triggering urgent genetic investigation.
Nephrocalcinosis	HP:0000121	Calcium deposits in kidney tissue documented on imaging; present congenitally since birth.
Short stature	HP:0004322	Length and height measurements significantly below expected percentiles for age and familial background.
Failure to thrive	HP:0001508	Repeatedly documented poor weight gain and inadequate linear growth across paediatric records.
Skeletal muscle atrophy	HP:0003202	Generalized reduction in muscle bulk and muscular development observed alongside growth failure.
Premature birth	HP:0001622	Preterm delivery at 32 weeks gestation.
Small for gestational age	HP:0001518	Severe low birth weight of approximately 1 kg, consistent with intrauterine growth restriction (IUGR).
Recurrent spontaneous abortion	HP:0200067	Maternal/parental history of multiple pregnancy losses prior to the proband's birth.

3. METHODOLOGY & QUALITY METRICS

Whole genome sequencing was conducted on high-molecular-weight genomic DNA. Paired-end sequencing reads were aligned to the human reference genome GRCh38. Variant detection was executed for single nucleotide variants and short insertions/deletions. Stringent analytical quality thresholds were enforced: read depth (DP ≥ 10), genotype quality (GQ ≥ 30), mapping quality (MQ ≥ 50), and filter status PASS. Variant consequences were annotated using Ensembl Variant Effect Predictor, ClinVar, gnomAD, CADD, SIFT, and PolyPhen-2. ACMG/AMP 2015 germline criteria and CPIC pharmacogenomic guidelines were applied for interpretation.

4. GENOMIC FINDINGS

Primary Candidate: Mosaic Variegated Aneuploidy Syndrome 1 (BUB1B)

BUB1B Compound Heterozygous Candidate (Estimated Causal Probability = 1.0) AUTOSOMAL RECESSIVE

Two rare variants were identified in *BUB1B* (NM_001211.6 / ENST00000287598.11), separated by 10.9 kb on chromosome 15q15.1. Both demonstrate high sequencing quality with balanced allelic depth and GQ 99.

Coordinates	cDNA / Protein	Zygosity & Quality	gnomAD AF	In Silico Scores	ClinVar	ACMG Classification
chr15:40209701 T>G rs759242053	c.2210T>G **p.Leu737Ter**	Heterozygous (0/1) AD: 21/25, DP: 46, GQ: 99	2.78×10^{-5} (0 homozygotes)	CADD: 36.0	Pathogenic / Likely Pathogenic (533901)	PATHOGENIC PVS1, PM2, PP3 (Score: 10)
chr15:40220612 T>G rs966509737	c.3006T>G **p.Asn1002Lys**	Heterozygous (0/1) AD: 15/13, DP: 28, GQ: 99	Absent (0.000)	SIFT: 0.01 (Damaging) PolyPhen: 0.997 (Damaging)	VUS (4600147)	VUS (LEANING PATHOGENIC) PM2, PP3

Note: Short-read WGS lacks physical phasing across 10.9 kb (PGT/PID unassigned). Parental segregation is required to confirm trans configuration.

Candidate Phenotype Modifier: Congenital Nephrocalcinosis

SLC34A1 (chr5:177386432 C>T) - Solute Carrier Family 34 Member 1

VUS / MODIFIER

Coordinates	cDNA / Protein	Zygosity & Quality	gnomAD AF	Predictions	ClinVar	Clinical Correlation
chr5:177386432 C>T	c.398C>T **p.Ala133Val**	Heterozygous (0/1) AD: 24/25, DP: 49, GQ: 99	0.0101 (1.0%)	SIFT: Deleterious (0.0)	Conflicting interpretations	Renal sodium-phosphate cotransporter (NaPi-IIa). Candidate contributor to hypercalciuria and nephrocalcinosis.

Secondary & Incidental Findings

Gene	Variant	Zygosity	gnomAD AF	ClinVar & ACMG	Clinical & Therapeutic Implications
ATM	chr11:108271261 T>C c.2932T>C (p.Ser978Pro)	Het (0/1) DP: 37, GQ: 99	0.00495	VUS / Other	DNA damage response modifier. Modulates cellular sensitivity to ionizing radiation and replication-stress-inducing agents.
GNRHR	chr4:67753920 C>T c.416G>A (p.Arg139His)	Het (0/1) DP: 60, GQ: 99	0.00095	PATHOGENIC	Autosomal recessive hypogonadotropic hypogonadism 7. Asymptomatic carrier state (no second allele identified).
FLG	chr1:152307547 G>A c.7339C>T (p.Arg2447Ter)	Het (0/1) DP: 38, GQ: 99	0.00766	PATHOGENIC	Common loss-of-function allele associated with increased susceptibility to atopic dermatitis and ichthyosis vulgaris.
MC3R	chr20:56249470 CAT>C c.628_629del (p.Met210ValfsTer101)	Het (0/1) DP: 36, GQ: 99	0.00021	VUS / High Impact	Rare frameshift in melanocortin 3 receptor. Linked to altered energy homeostasis and linear growth kinetics.

Pharmacogenomic Profile (PGx)

Germline variants in core drug-metabolizing enzymes and transporters were evaluated:

Gene	Defining Variant	Genotype	Predicted Phenotype	Actionable Drug Implications
CYP3A5	chr7:99672916 T>C rs776746 (c.219-237A>G)	Homozygous Alt (1/1) ***3/*3**	NON-EXPRESSOR / POOR METABOLIZER	Absence of functional CYP3A5 shifts all CYP3A oxidative clearance to CYP3A4. Heightened risk of toxicity when CYP3A4 inhibitors are co-prescribed.
CYP2D6	chr22:42128945 C>T rs3892097 (c.353-1G>A)	Heterozygous (0/1) ***4 allele carrier**	INTERMEDIATE METABOLIZER	Decreased metabolic activation of codeine and tramadol; possible altered antiemetic efficacy with ondansetron.
CYP2C19	rs4244285 (*2, null) rs12248560 (*17, increased)	Compound Heterozygous ***2/*17**	INTERMEDIATE METABOLIZER	Balanced intermediate metabolic capacity for CYP2C19 substrates. Normal clearance profile expected for secondary pathways.

5. CLINICAL INTERPRETATION & BIOLOGICAL MECHANISM

Molecular Mechanism of BUB1B Deficiency

The patient's clinical constellation matches **Mosaic Variegated Aneuploidy Syndrome 1 (MVA1, OMIM #257300)**. BUB1B encodes BUBR1, a core kinase and scaffolding component of the Mitotic Checkpoint Complex (MCC) alongside MAD2, BUB3, and CDC20. In normal mitosis, unattached or misaligned kinetochores recruit BUBR1 to assemble the MCC, which binds and inhibits the Anaphase-Promoting Complex/Cyclosome (APC/C-CDC20), delaying anaphase onset until all sister chromatids achieve bipolar tension.

The observed genotype represents an uncoupled loss-of-function configuration: an unexpressed null allele (p.Leu737Ter, undergoing nonsense-mediated decay) combined with a hypomorphic missense allele (p.Asn1002Lys) situated in the C-terminal pseudokinase domain. This severely depletes functional BUBR1 protein. The downstream consequences follow two distinct pathways:

- **Embryonal Tumorigenesis (Rhabdomyosarcoma):** Spindle assembly checkpoint (SAC) incompetence allows premature chromatid separation (PCS) and accelerated anaphase entry. Chronic chromosome missegregation produces high-grade chromosomal instability (CIN) and variegated mosaic aneuploidy, driving rapid loss of heterozygosity at tumor suppressor loci and fueling embryonal rhabdomyosarcoma.
- **Somatic Progeroid Decline (IUGR, Muscle Wasting, Failure to Thrive):** In non-transformed progenitor cells, persistent chromosome missegregation triggers chronic DNA damage cascades, driving cellular senescence through p16INK4a activation. This causes early stem-cell exhaustion, growth restriction, and severe skeletal muscle atrophy.

Genotype-Phenotype Correlation for Congenital Nephrocalcinosis

Nephrocalcinosis is not a recognized manifestation of MVA1. Screening of 47 renal tubulopathy loci revealed a heterozygous missense variant in *SLC34A1* (p.Ala133Val, AF 1.0%). While recessive biallelic SLC34A1 mutations cause hypophosphatemic rickets and nephrocalcinosis, monoallelic SLC34A1 variants are established susceptibility factors for

idiopathic hypercalciuria and renal calcification. This finding represents a putative phenotypic modifier rather than an independent monogenic diagnosis.

Repositioned Drug Candidates & Dosing Framework

> **Therapeutic Strategy:** Small-molecule replacement of lost BUBR1 protein is biologically unachievable. Rational drug repositioning targets the collateral vulnerabilities created by SAC collapse, proteotoxic stress, and cellular senescence.

Candidate Drug	Target & Repurposing Rationale	Clinical Dosing Framework	Patient-Specific Genomic & Organ Adjustments
Bortezomib (Proteasome Inhibitor)	Targets severe proteotoxic stress resulting from stoichiometric protein imbalances in aneuploid cells. BUBR1-deficient tumors depend heavily on proteasome degradation for survival.	**1.2 to 1.3 mg/m² per dose** IV push or SC on days 1, 4, 8, 11 of a 21-day cycle (COG ADVL0916).	**Renally Safe:** Hepatically cleared; safe in congenital nephrocalcinosis. Carfilzomib is strictly avoided due to severe renal toxicity risks. Subcutaneous route preferred to minimize neuropathy.
Azenosertib (ZN-c3) (Clinical-Stage WEE1 Inhibitor)	Aneuploid cells experience intense replication stress and depend on WEE1-mediated G2/M arrest. WEE1 inhibition abrogates the G2 checkpoint and forces cells prematurely into mitosis, triggering mitotic catastrophe.	Active Phase 1/2/3 oral WEE1 inhibitor in clinical development for solid tumors and sarcomas (NCT04158336). Pediatric dosing requires BSA-based escalation.	**ATM & CYP3A Safeguards:** Co-occurring ATM p.Ser978Pro heightens cellular sensitivity to DNA replication stress; initiate at a conservative dose level. Close monitoring required given the patient's CYP3A5*3/*3 non-expressor status.
Navitoclax (ABT-263) (Senolytic BCL-2/BCL-xL Inhibitor)	Selectively clears p16INK4a-positive senescent progenitor cells driving skeletal muscle atrophy and failure to thrive in BUBR1 deficiency.	**Intermittent Pulsed Regimen:** 25 to 50 mg/m² orally for 1 to 2 consecutive days every 14 to 28 days.	**Thrombocytopenia Mitigation:** Continuous oncologic exposure is contraindicated due to platelet toxicity. Intermittent pulsing clears senescent cells while sparing megakaryocytes.

6. CLINICAL & DIAGNOSTIC RECOMMENDATIONS

1. **Parental Phasing Analysis:** Targeted Sanger sequencing of both biological parents for *BUB1B* c.2210T>G and c.3006T>G to confirm compound heterozygosity in trans.
2. **Cytogenetic PCS/MVA Karyotyping:** Specialized cytogenetic analysis on peripheral blood lymphocytes assessing for premature chromatid separation (PCS) and mosaic aneuploidy across ≥50 metaphases. Confirmation provides formal phenotypic validation of MVA1.
3. **Pediatric Oncologic Surveillance:** Children with MVA1 face elevated risks of secondary embryonal malignancies. Protocol-driven surveillance is recommended, including abdominal ultrasound screening for Wilms tumour every 3 months until age 8 years.
4. **Nephrology & Metabolic Evaluation:** Assessment of congenital nephrocalcinosis via renal ultrasound, urinary calcium-to-creatinine ratio, serum phosphate, tubular reabsorption of phosphate (TmP/GFR), and parathyroid hormone levels.

5. **Genetic Counseling:** Recurrence risk counseling for the family. Confirmed biallelic inheritance carries a 25% recurrence risk for MVA1 in future pregnancies, which may contextualize the parental history of recurrent spontaneous abortions.

7. SCIENTIFIC REFERENCES

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8. TECHNICAL & METHODOLOGICAL LIMITATIONS

- **Unphased Short-Read Sequencing:** The physical distance between the two BUB1B variants is 10.9 kb. Short-read sequencing cannot determine physical phase. Trans configuration is inferred from ultra-low population frequencies.
- **Structural Variant Detection:** Standard short-read WGS has limited sensitivity for complex structural rearrangements, deep intronic regulatory variations, and repeat expansions.
- **In Silico Classifications:** Predictive algorithms (CADD, SIFT, PolyPhen) do not substitute for direct functional cellular assays.
- **Research Scope:** Dosing guidance is synthesized from published oncology trials and preclinical senolytic models, and must not supersede the individualized care plan of an attending paediatric oncologist.